ISSN 1069-3513, Izvestiya, Physics of the Solid Earth, 2018, Vol. 54, No. 1, pp. 163–177. © Pleiades Publishing, Ltd., 2018.
Original Russian Text © A.G. Iosifidi, V.A. Mikhailova, V.V. Popov, E.S. Sergienko, A.V. Danilova, N.M. Otmas, 2018, published in Fizika Zemli, 2018, No. 1, pp. 172–186.

The Carboniferous of the Moscow Syneclise: Paleomagnetic Data

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Received March 3, 2017

Abstract—New data for the Early and Late Carboniferous sections of the Russian platform (Moscow syneclise and Donbass) are presented. Magneto-mineralogical studies are carried out to identify the magnetic minerals—carriers of natural remanent magnetization. Extensive Late Paleozoic remagnetization of Carboniferous rocks is revealed. The obtained paleomagnetic data allowed us to determine the average paleomagnetic poles for the Gzhelian, Serpukhovian, and Visean stages of Carboniferous deposits of the Moscow syneclise.

Keywords: Carboniferous deposits, paleomagnetic poles, Russian Platform

DOI: 10.1134/S1069351318010081

INTRODUCTION

Extensive paleomagnetic studies of Carboniferous deposits within the Russian platform were conducted in the late 1950s and the beginning of the 1960s in the Paleomagnetic laboratory of the All-Russia (formerly USSR) Petroleum Research Exploration Institute (VNIGRI) under the guidance of A.N. Khramov (Khramov and Komissarova, 1963; Khrramov, 1963a; 1963b; 1967; *Paleomagnetizm*..., 1974). Practically all the known sections of Carboniferous red rocks of the Moscow syneclise were sampled and investigated. Reinvestigation a part of the old collections allowed us to estimate the reliability of some paleomagnetic determinations of the 1960s (Iosifidi and Mikhailova, 2017). The analysis of the existing data for the Carboniferous deposits of the East European platform (EEP) that are presented in the Global Paleomagnetic Database (GPMDB) (McElhinny and Lock, 1996; Pisarevsky, 2005) and published in the literature has shown that the number of determinations for the Early Carboniferous (with a duration of ~40 Ma) that meet the present-day reliability criteria is half the number of the determinations for the epochs of the Late Carboniferous (~18 Ma). This situation prevents us from correctly estimating the Carboniferous kinematics of the EEP's motion. The reliability of the paleomagnetic data increases in the cases of the identification of bipolar components. The presence of zones with normal and reversed magnetization polarity makes it possible to perform the reversal test, a powerful instrument for revealing the synchronism of the normally and reversely magnetized rocks. These determinations have a key value and allow us to estimate the correctness of the results derived from the unipolar NRM components. Another difficulty is associated with the question of remagnetization of the rocks during the Kiama hyperchron.

The present work continues the studies of the Carboniferous rocks of the Russian Platform (reinvestigation the old collections) which were started in (Iosifidi and Mikhailova, 2017). In this paper we present the results that rely on the collections that were acquired by Khramov and other geophysicists of the Paleomagnetic laboratory of VNIGRI in the 1950s–1970s from the Carboniferous rocks of the Russian platform and a collection of rocks sampled by paleomagnetologists from St. Petersburg State University and VNIGRI in 2014, from the key sections of the Visean and Serpukhov stages in the basin of the Msta River.

STUDY OBJECTS AND EXPERIMENTAL TECHNIQUE

Moscow region. The collection of Carboniferous rocks from the Shchelkovskaya layers of the Gzhelian stage was acquired in the vicinity of Noginsk (Bun'kovskii quarry) (the collection of R.A. Komissarova, 1962). In this paper we study the tile-red clays occupying 3 m of the section and covering 5 stratigraphic levels (*Geologiya*..., 1971). Subhorizontal bedding and sampling coordinates 55.83° N, 38.60° E (Figs. 1, 2).

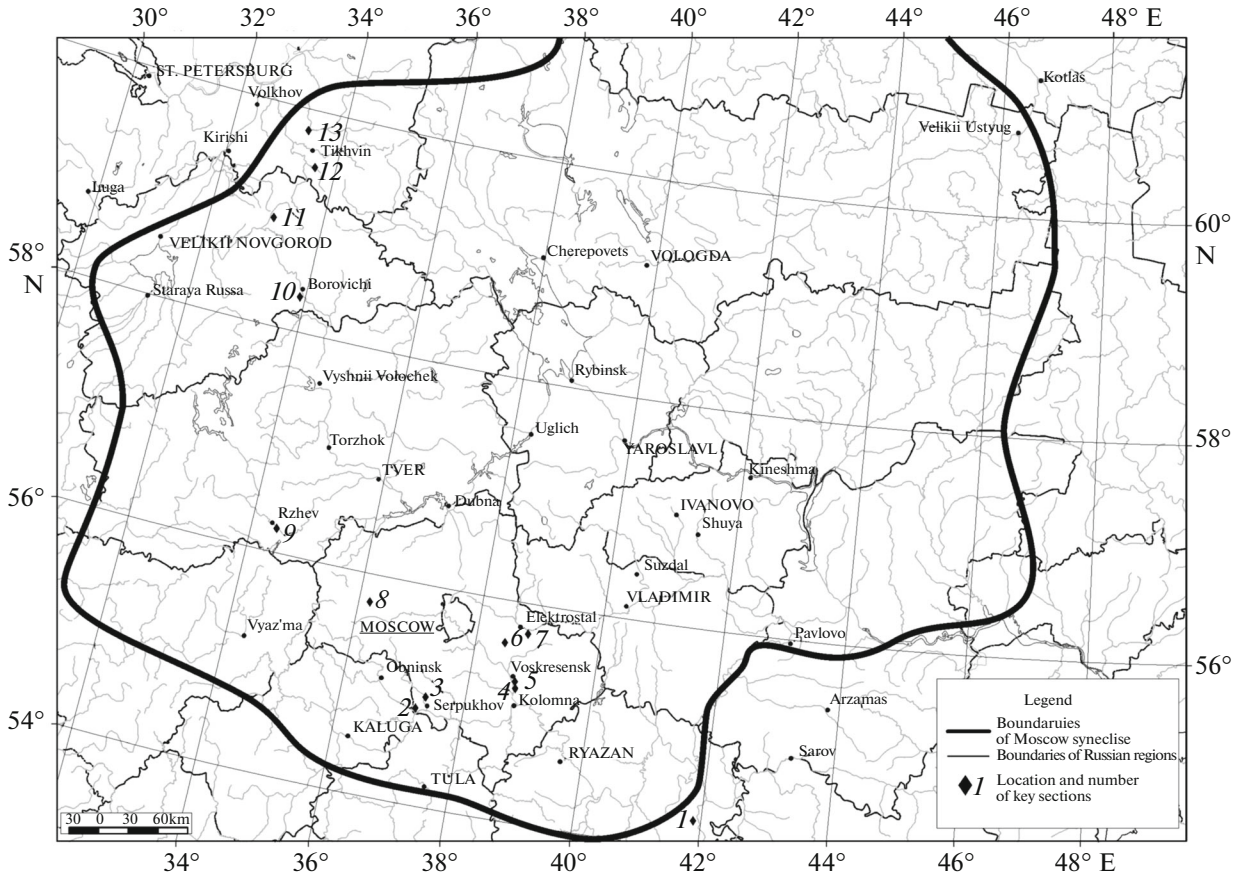


Fig. 1. Schematic map of natural Carboniferous outcrops of Moscow syneclise: (*1*) town of Shatsk; (*2*) Nara River; (*3*) Skniga River; (*4*) Peski station; (*5*) Gigant quarry; (*6*) Gzhel station; (*7*) Bun'kovskii quarry; (*8*) Protva River; (*9*) Volga River; (*10*) Msta River; (*11*) Priksha River; (*12*) quarry of mine no. 13; (*13*) Lininka River.

The rocks of the Kasimovian stage of the Carboniferous (Voskresenskaya stratum) are sampled in the region of the town of Voskresenk (*Gigant* plant quarry, lower reaches of Moskva River), the collection of Komissarova, 1962. The sampled section has a total thickness of 5.5 m and includes nine stratigraphic levels (*Geologiya*..., 1971). Subhorizontal bedding, sampling coordinates 55.26° N, 38.75° E (Figs. 1, 2).

Novgorod region. The collection was sampled by the paleomagnetologists of VNIGRI and St. Petersburg State University in 2014 from the Visean and Serpukhovian reference sections in the vicinity of the town of Borovichi, Novgorod region, in the Msta river basin (Savitskii et al., 2012). The Visean is represented by the rocks of the Egol'skaya group of the Venevskii horizon (limestones, red brown clays); the Serpukhovian is composed of limestones and flints of the Rovenskaya group of the Tarussian horizon and dolomitic limestones of the Poneretskaya group of Steshevskii horizons. The rocks are sampled from six outcrops. The total thickness of the sampled interval is 18 m and 47 stratigraphic levels are covered (Figs. 1 and 2). Subhorizontal bedding, average sampling coordinates 58.3° N, 34.0° E.

The section of the Priksha River (in the vicinity of Sherekhovichi village) is a sandy clay stratum of the Tulskian horizon of the Visean stage. The thickness is 21 m, 14 stratigraphic layers (collections of Khramov (1959) and Pogarskaya and Komissarova (1968)), Figs. 1 and 2. Subhorizontal bedding, sampling coordinates 58.3° N, 33.6° E.

Donbass. The region of the Sukhoi Yar village (north of the town of Gorlovka and Zaitsevo village). Sandy clay stratum in the basal part of the Kartamyshskaya group (cupriferous sandstone group) related to the Gzhelian stage of Late Carboniferous (Davydov and Khramov, 1991; *A Geological*..., 1960). Total thickness 60 m, 23 stratigraphic levels (Collection of Khramov, 1960). Bedding elements vary within the following limits: dip azimuth 16°–29°, dip angle 32°–56°; sampling coordinates 48.4° N, 38.0° E.

The laboratory paleomagnetic studies and processing of the obtained results followed the standard procedure (Khramov et al., 1982; Khramov, 1987). Thermal demagnetization of the rocks was carried out with the use of the equipment placed in five-layer permalloy shields (original design by VNIGRI) and the ther-

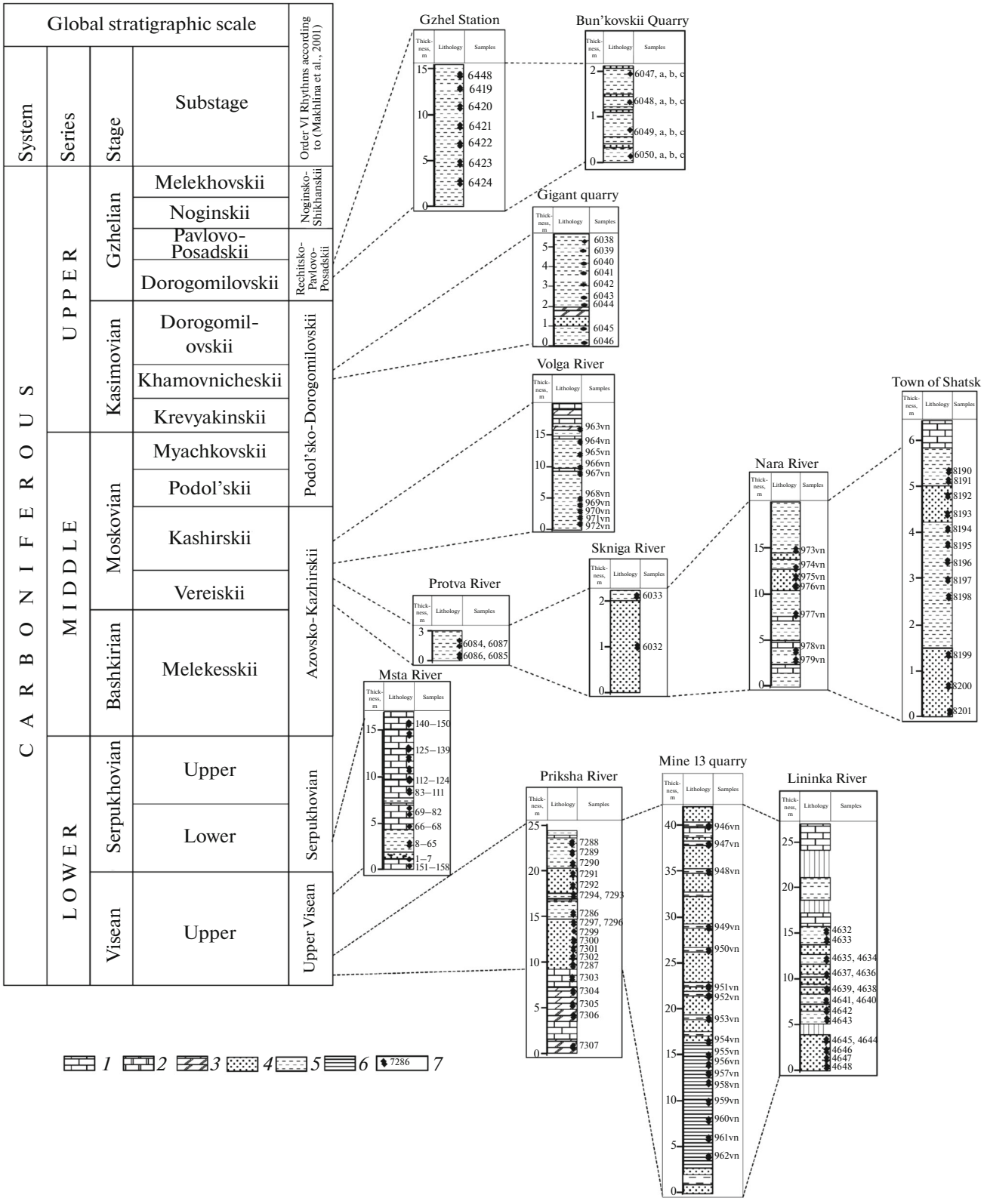


Fig. 2. Scheme of correlation of Carboniferous sections of Moscow syneclise to general magnetostratigraphic scale. Rock types: (*1*) limestones; (*2*) dolomites; (*3*) dolomitized marls; (*4*) sandstones; (*5*) clays; (*6*) bauxites; (*7*) sampling sites.

Table 1. Range of variations of J_n and K_m in studied rocks

No.	Rock age	Number of samples	Range of variations of J_n, mA/m	Geometric mean J_n, mA/m	Number of samples	Range of variations of K_m, 10^{-6} SI units	Geometric mean K_m, 10^{-6} SI units
1	C_3gz	23	1.7–46.7	9.6	11	164–286	212
2	C_1v Priksha R.	58	0.1–50.7	6.2	38	27–354	165
3	C_1s Msta R.	66	0.02–3.49	0.10	25, limestones	1–181	22
4	C_1v Msta R.	55	0.01–45.90	2.50	12, clays	121–186	160

mal demagnetizer TD48 (USA). NRM was measured by JR-4 and JR-5 magnetometers (AGICO, Czech Republic) and the SRM-755 SQUID magnetometer (United States). Magnetic susceptibility measurements were conducted by KLY-3 kappabridge (AGICO, Czech Republic) and the analysis of the temperature dependence of the magnetic susceptibility on a MFK-1-FA kappabridge (AGICO, Czech Republic). The microscopic studies, besides visualization of ore minerals, included microprobe analysis and, when necessary, obtaining the charts of the distributions of the chemical elements. The works were conducted at the facilities of the Research Park of St. Petersburg State University: the Center for Geo-Environmental Research and Modeling (GEOMODEL) and the Resource Center and Interdisciplinary Resource Center for Nanotechnology. Specimens for microscopic analysis of limestones were prepared in the form of polished thin sections. It was not possible to produce thin sections from the samples of red-brown clays and isolate the magnetic phase into a separate specimen. Therefore, the following procedure was used. A clay sample was ground into powder and placed on adhesive tape after which the selected area (~75 × 50 μm) was scanned, and the integral cumulative pattern of the distribution of elements was constructed. Specifically, the main rock-forming minerals Fe, O, Si, and Al were selected. Ore minerals were identified as the zones of joint clusters of iron and oxygen. For the analysis of the obtained data and visualization of the results, we used the program packages of (Popov, 2007; 2008) and (Enkin, 1994). The reversal test and consistency test were performed in the modification (McFadden and McElhinny, 1990). In data processing, the statistics were calculated on the sample level (stratigraphic levels).

EXPERIMENTAL DATA

Magnetic properties. The values of the scalar magnetic characteristics (NRM (J_n) and magnetic susceptibility (K_m)) in the studied rocks vary widely (Table 1). The analysis of the temperature dependences of magnetic susceptibility and the interpretation of the observed features show the following facts. In the samples from the collection of the Shchelkovskaya stratum (the Gzhelian stage of the Carboniferous) (Fig. 3a, sample *6049*), one magnetic phase with the Curie point of 560–580°C probably corresponding to nonstoichiometric magnetite is isolated.

In the samples of red-brown clays of the Venevskii horizon (Egol'skaya suite, Msta River), one magnetic phase with the Curie points close to hematite (650–690°C) is established. A typical curve (sample *052*) is shown in Fig. 3b. On cooling, magnetite is restored. It manifests itself as the Verwey peak in the low temperature part of the curve of repeated heating.

In the collection of limestones from the same outcrop (the Stashevskian and Tarussian horizons), magnetic measurements were carried out for 6–8 samples per horizon. The Curie points are 680–690°C (Fig. 3c, sample *115*).

The results of the microscopic studies for the limestone samples from the reference Carboniferous section sampled in the vicinity of Egla village (Msta River) are illustrated in Fig. 4. Hematite grains of structures, size, and probably of different generations are detected. Firstly, it is the isolated oolite-like grains with a size of 1–20 μm and their rounded clusters probably developing along the contours of organic remains. Secondly, it is the large mineral aggregates reaching hundreds of μm in size. They were probably formed as a result of the merging of fine grains. Figures 5a–5e show the results of chemical element mapping and the image of an isolated ore mineral grain for samples of red-brown clay from the same outcrop section. We used this method only because it was not possible to prepare polished thin sections and extract the ore fraction. In these samples, the analysis revealed hematite in the form of fine oolite-like (5–8 μm) grains. The image of the surface of the isolated grain and the results of microanalysis which correspond to hematite are illustrated in Fig. 5f. The hematite grain is coated with a clay shell. The microprobe analysis detects

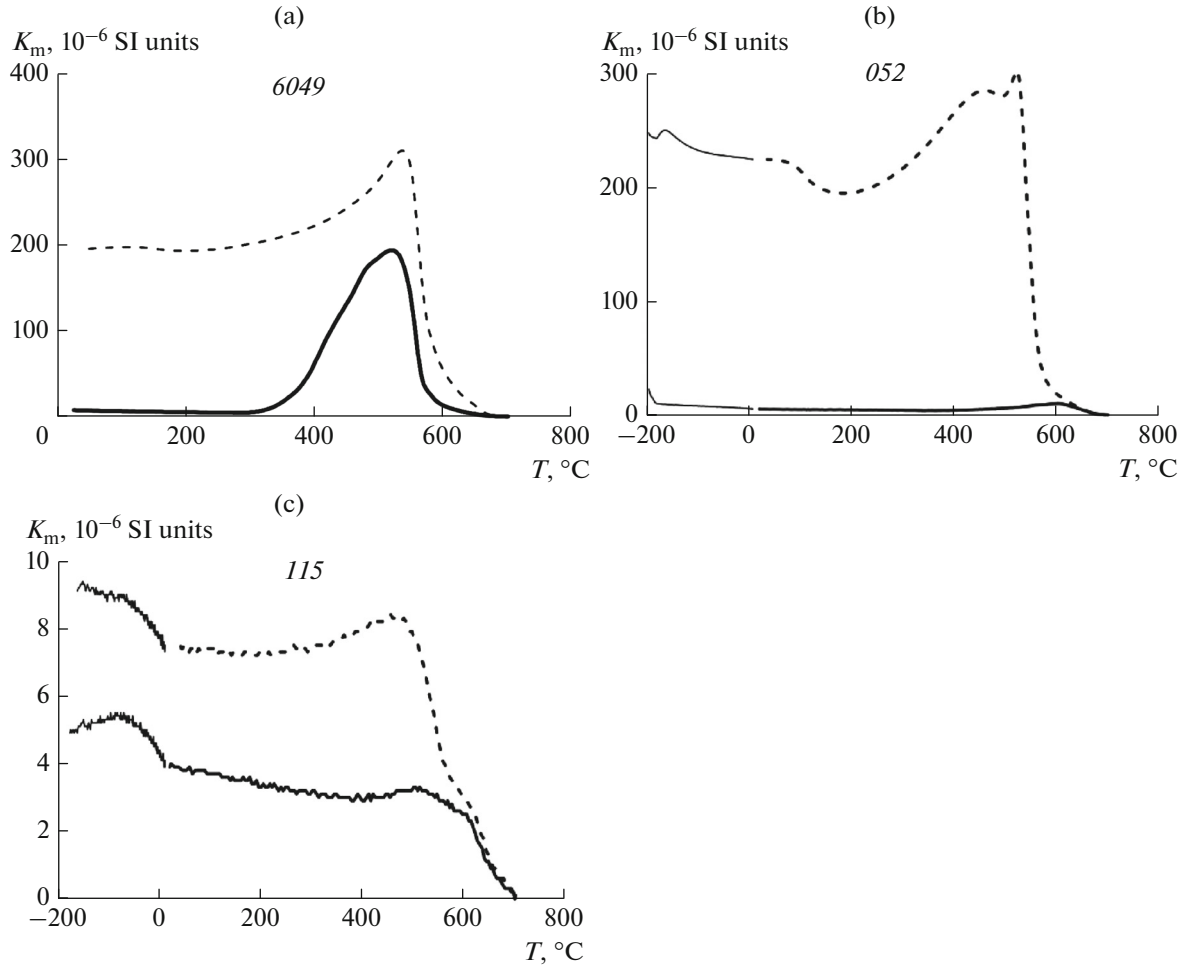


Fig. 3. Examples of thermomagnetic analysis of Carboniferous rocks: temperature curves of magnetic susceptibility under heating (solid line) and cooling (dashed lines). Samples *6049* (clay, Bun'kovskii quarry), *052* (clay, Msta River), *115* (limestone, Msta River).

minor amounts of aluminum and silicon; and the amount of oxygen is high.

The structural features of hematite are very diverse and depend on the conditions in which hematite is formed. The minimal temperature of hematite formation is 200–300°C in the laboratory conditions; however, in nature its crystallization is observed at even lower temperatures. Hematite is most frequently formed from the martitization of magnetite but also occurs as an independent (primary) mineral, i.e., a mineral that crystallizes during the formation of a rock rather than due to secondary transformations.

In sediments, this mineral can be formed during the diagenesis processes or under the load metamorphism. Thin-plate, flaky, fine-grain hematite (according to the terminology of P. Ramdohr (Ramdohr, 1962)) most frequently occurs in rocks of hydrothermal origin. In these conditions, hematites can also be finely dispersed or occur in the form of rounded grains without distinct boundaries (oolite-like, as in our case). Large grains are formed, as a rule, as a result of recrystallization. The paleogeographical setting initially of the time of formation of clay deposits and subsequently the studied limestones (Visean/Serpukhovian boundary and the beginning of the Serpukhovian Age) is marked by a change of the predominantly shallow sea with the accompanying aridization of the climate and a tectonic rearrangement with the formation of deep sedimentation zones (Makhlina et al., 1993; Osipova and Vel'skaya, 1967). These conditions can promote the growth of large mineral aggregates of fine hematite grains. Thus, the primary (at least, the earliest formed) magnetic mineral in the studied rocks of the Msta River is fine-grain hematite, and it is this hematite that contains information about the most ancient magnetization component. The mineral aggregates established in limestones are most likely to be associated with a more recent magnetization.

Component analysis of J_n. The magnetization components J_n were detected by stepwise thermal demagnetization and demagnetization by an alternating magnetic field.

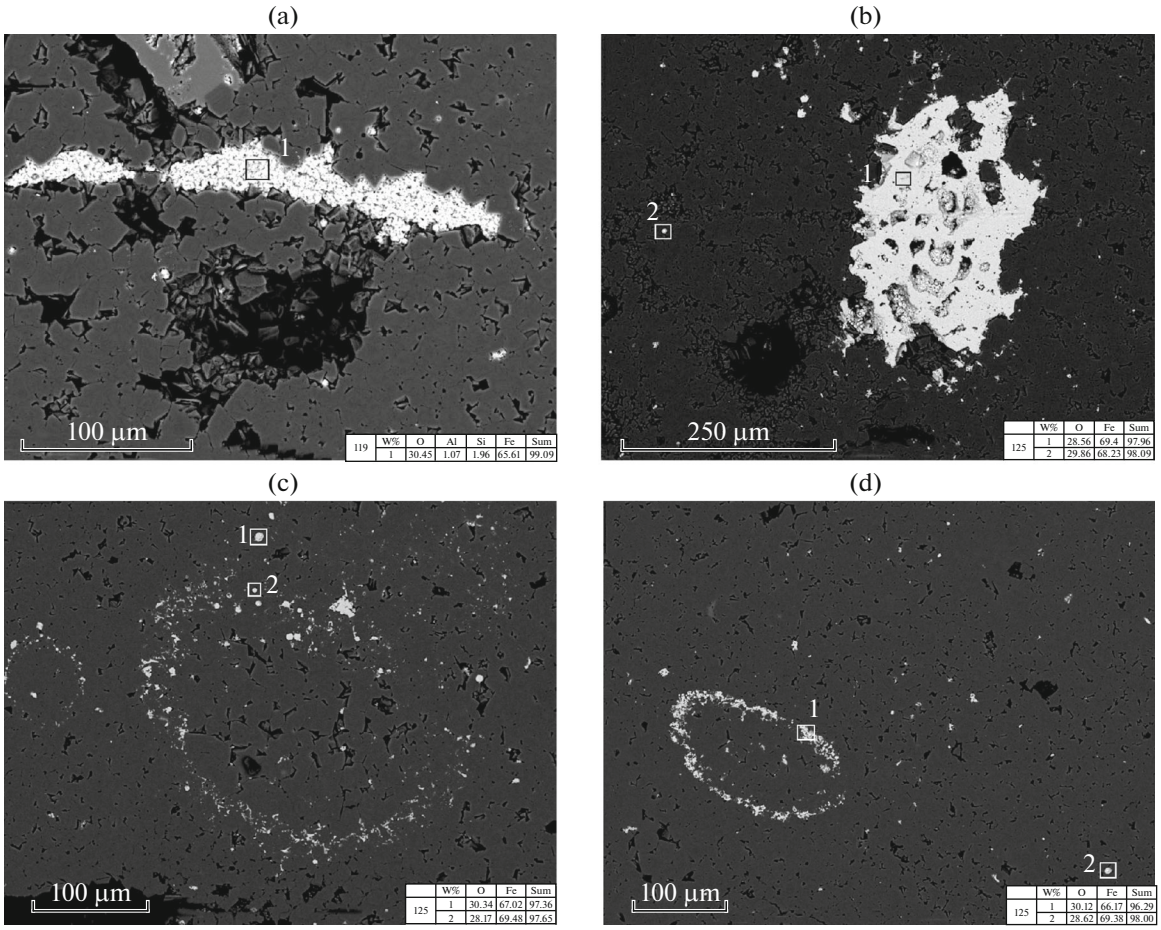


Fig. 4. Microphotographs of polished thin sections and results of micoranalysis for limestones of reference Carboniferous section sampled in region of Egla village (Msta River). (a), (b), Large mineral aggregations of hematite up to hundreds of μm and oolite-like grains up to 20 μm; (c), (d) clusters of hematite grains of 1–10 μm and oolite-like isolated grains up to 20 μm. Areas delineated by boxes mark zone of microprobe analysis.

Among the rocks of the Gzhelian age, there are samples with single-component J_n and two-component J_n in the intermediate- and high-temperature demagnetization intervals, as shown by the Zijderveld plots (Fig. 6, samples *6048-2v* and *6050-2v*). The temperature interval of the isolation of the C3 component lies within 200–560° C. Component C0 is isolated in the interval 460–540° C. The direction of the C0 component differs from component C1 both in declination and inclination. The statistical characteristics of the identified components are presented in Table 2, and their distributions are shown in Figs. 7a and 7b.

In the rocks of the Kasimovian Stage, a single J_n component is isolated. The intermediate-temperature C3 component is isolated in the interval 200–540°C (Fig. 6, sample *6045-1*). The distribution of component C3 for the rocks of the Krevyakinskii horizon (Voskresenskaya stratum) of the Kasimovian Stage is shown in Fig. 7c and the statistical characteristics are presented in Table 2.

The Visean and Serpukhovian rocks (Msta River) have three J_n components. The first component, A, is identified in clays (Fig. 6, sample *014*) in the temperature interval 100–300°C. In some limestones this component is isolated in the interval 200–675°C, which is due to the complete remagnetization by the present geomagnetic field. The second component P is isolated in the temperature intervals 200–675°C, and in the case of demagnetization by an alternating field, it is recorded in the amplitude interval of 5–50 mT (Fig. 6, sample *014*). Component C1 is isolated in the high-temperature interval 200–675°C (Fig. 6, sample *071-1*). In some limestones, chemical alterations are observed after heating to 350–450°C. Attempts to apply the combined demagnetization (Fig. 6, sample *M115-4*) failed. In these cases, in the intermediate-temperature interval, the analysis sometimes reveals component C1. This component is established in the Serpukhovian and Visean deposits of the Msta River in several outcrops, and its distribution is different from Fisher's distribution. Unfortunately, the bedding elements of the

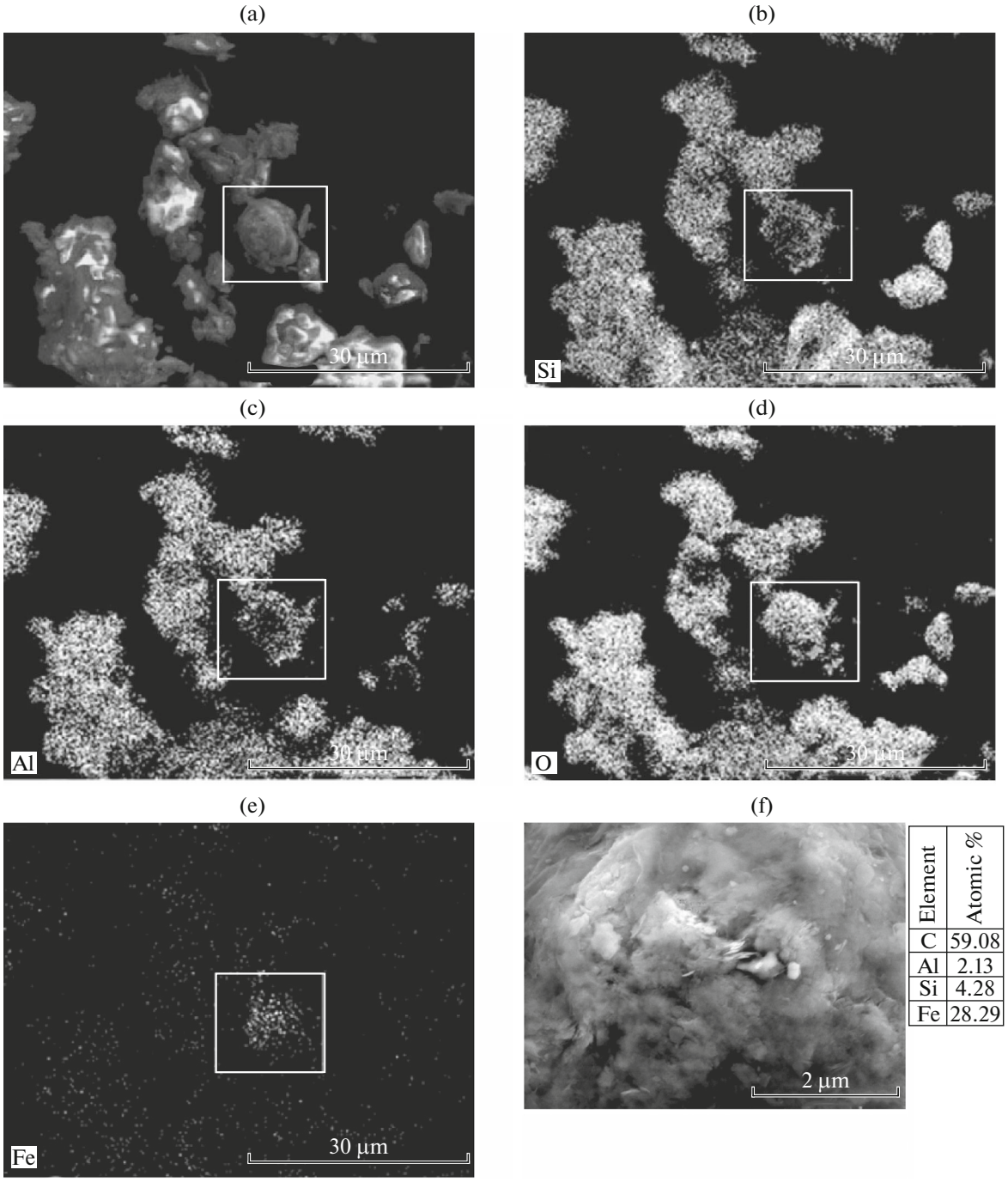


Fig. 5. Results of microscopic studies of red-brown clay samples of reference Carboniferous section sampled in region of Egla village (Msta River). (a), microphotograph of initial area for mapping chemical elements. Distribution maps of elements: (b) silica, (c) aluminum, (d) oxygen, (e) iron; (f) microphotograph of individual hematite grain coated with clay shell.

strata in the sections of the Msta River differ insignificantly (by a few degrees), which prevents efficient application of the method of remagnetization planes (Khramov, 1987; 1996; Khramov et al., 1961, 1982, 1987). For estimating the direction of component C1, we used the new modification of the method of remagnetization planes suggested by Khramov, namely, the method of "sliding normals" to the remagnetization circles (Khramov et al., 2011). All the obtained directions of component C1 averaged over the stratigraphic levels are illustrated in Fig. 7h. Examples of determining the positions of the sliding normals (the circles through the internal areas were drawn in a stratigraphic sequence by three points) are also shown. The final result is presented in Table 2, line 10, and shown in Fig. 7i. The distribution of components A, P, and C1

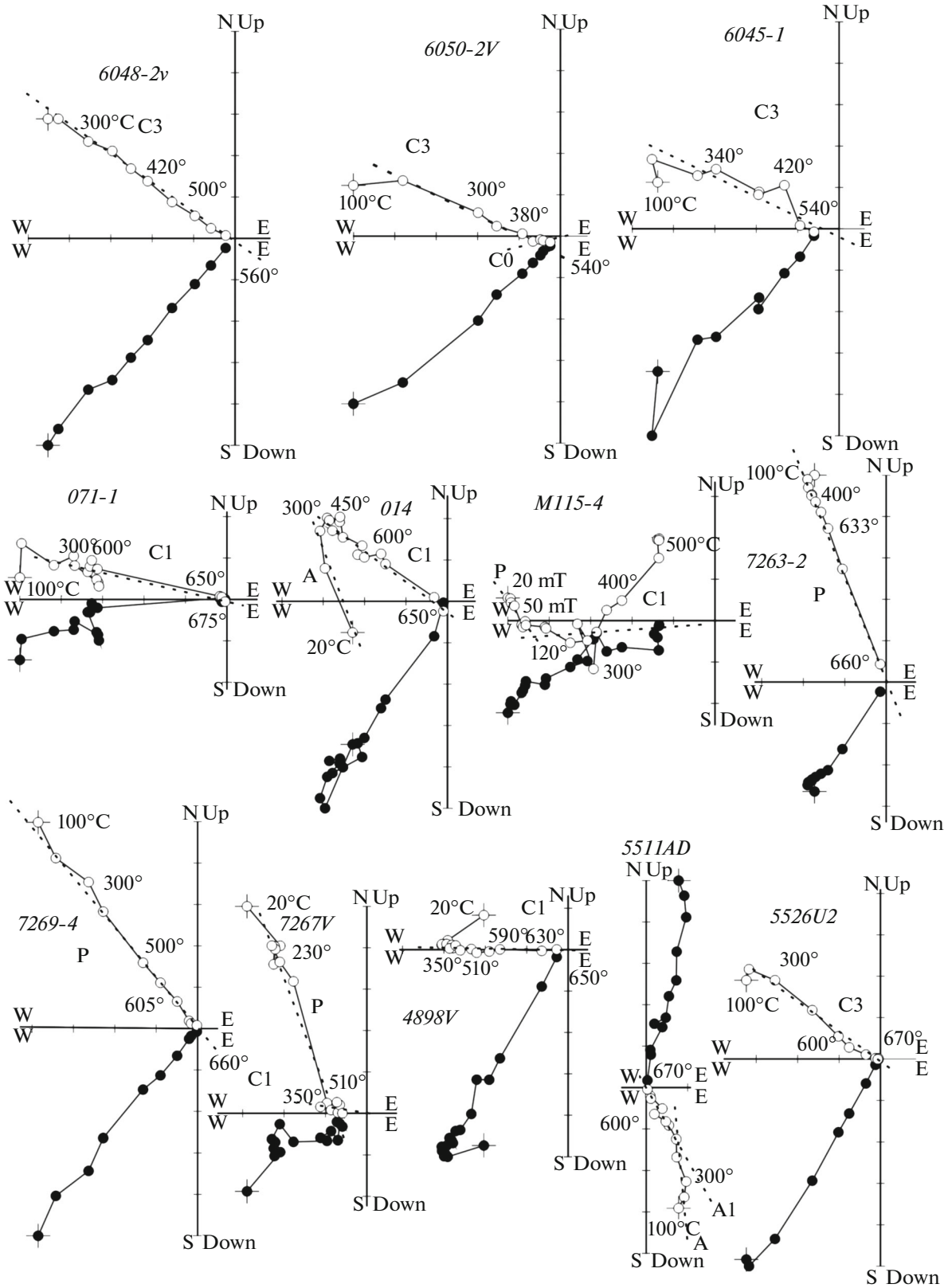


Fig. 6. Examples of stepwise thermal demagnetization of Carboniferous samples. Samples *6048-2V*, *6050-2V* are from Gzhelian; *6045-1*, from Kasimovian; *M115-4* from Serpukhovian; *014*, *071*, *7263-2*, *7269-4*, *7267V*, *4898V* from Visean stages of Carboniferous; samples *5511AD*, *5526U2* are from basal parts of Kartamysh suite, Donbass. Zijderveld plots are presented in stratigraphic coordinate system; open and filled circles are vector projections on vertical and horizontal plane, respectively. Dashed line shows segments of identification of J_n components; designations of component correspond to Table 2. Zijderveld plots are constructed with use of Remasoft 3.0 program (Chadima and Hrouda, 2006).

Table 2. Paleomagnetic directions and paleomagnetic poles for studied Carboniferous deposits

No.	Rock age	J_n comp. (isolation intervals, °C)	S.C.	N/n	$D°$	$I°$	K	α_{95}°	Φ,° N	Λ,° E	dp°	dm°	φ_m°
Moscow Region, Bun'kovski quarry, Shchelkovskii stratum, Gzhelian													
1	C_3g	**C3** (300–540°C)	s	5/10	225	–23	119	7	34	162	4	7	12
2		**C0** (460–540°C)	s	4/4	240	–3	20	21					
	Gigant plant quarry, lower reaches of Moscow River, basal part of Kasimovian												
3	C_3k	**C3** (300–540°C)	s	3/5	231	–24	206	9	32	154	5	10	13
Novgorod region, Serpukhovian (Steshevskii horizon, Poneretskaya suite and Tarusian horizon, Rovnenskaya suite)													
4	C_1s	**P** (20–700 mT)	s	11/11	217	–55	43	7	57	151	7	10	36
5		**C1** (30–400 mT)	s	4/4	223	–22	36	16	33	161	9	17	11
	Visean (Venevskii horizon, Egol'skaya suite)												
6	C_1v	**A** (100–300°C, 200–670°C)	g	40/40	356	59	6	10	71	224	11	15	40
7	C_1v	**P** (200–675°C)	s	15/24	217	–54	53	5	56	152	5	7	35
8		**P (4 + 7)**		26/35	217	–54	50	4	56	152	4	6	35
9		**C1** (300–650°C)	s	10/13	220	–21	32	9	34	168	5	10	11
10	C_1 s–v	m.s.n.	s	14	225	–20		11	31	160	6	12	10
	Priksha River, Tul'skii horizon, Visean												
11	C_1v	**P** (200–660°C)	s	3/6	39	51	155	10	53	152	9	14	32
12	C_1v	**P** (200–660°C)	s	30/55	218	–43	53	4	47	158	3	5	25
13		**C1** (300–660°C)	s	7/12	222	–15	97	6	30	164	3	6	8
	Donbass, sandy clay stratum in basal part of Kartamyshian suite												
14	C_3g	**A** (100–500°C)	g	8/8	348	67	52	8	82	322	11	13	50
15			s		9	33	66	7	59	201	5	8	18
16	C_3g	**A1** (650–675°C)	g	5/5	346	56	13	22	74	264			37
17			s		3	25	12	23	57	213			13
18	C_3g	**C3** (200–670°C)	g	10/10	225	–55	21	11					
19			s		214	–13	36	8	39	172	4	8	7
20	C_3g	m. s. n.	g	10/10	237	–71		6	55	94	9	11	55
21			s		213	–21		10	44	166	6	11	11

C.S. is coordinate system (g is geographic, s is stratigraphic); N is number of hand sample; n is number of specimens; $D°$, $I°$ are declination and inclination of average directions of magnetization components; K is precision parameter; α_{95}° is radius of confidence at 95% probability for average direction; Φ°, Λ° are latitude and longitude of paleomagnetic pole; dp°, dm° are semiaxes of confidence oval; φ_m° is paleomagnetic latitude, m. s. n. denotes that determination obtained by method of sliding normals to remagnetization circles (Khramov et al., 2011). Age of components: A is present-day, P is Permian, C3 is Late Carboniferous, C2 is Middle Carboniferous, and C1 is Early Carboniferous

for the rocks of the Serpukhovian and Visean ages is shown in Figs. 7d–7g; their statistical characteristics are presented in Table 2.

The Visean rocks (Priksha River) have three J_n components. The first component P is distinguished in clays (Fig. 6, samples *7263-2*, *7267V*, and *7269-4*) in the intervals 200–660 and 200–400°C. On one stratigraphic level, component P with normal polarity is identified (Table 2, Fig. 7j). The third component, C1, is isolated in the interval 300–660°C (Fig. 6, sample *4898V*), and in part of the samples in the interval 600–660°C. The distribution of components P and C1

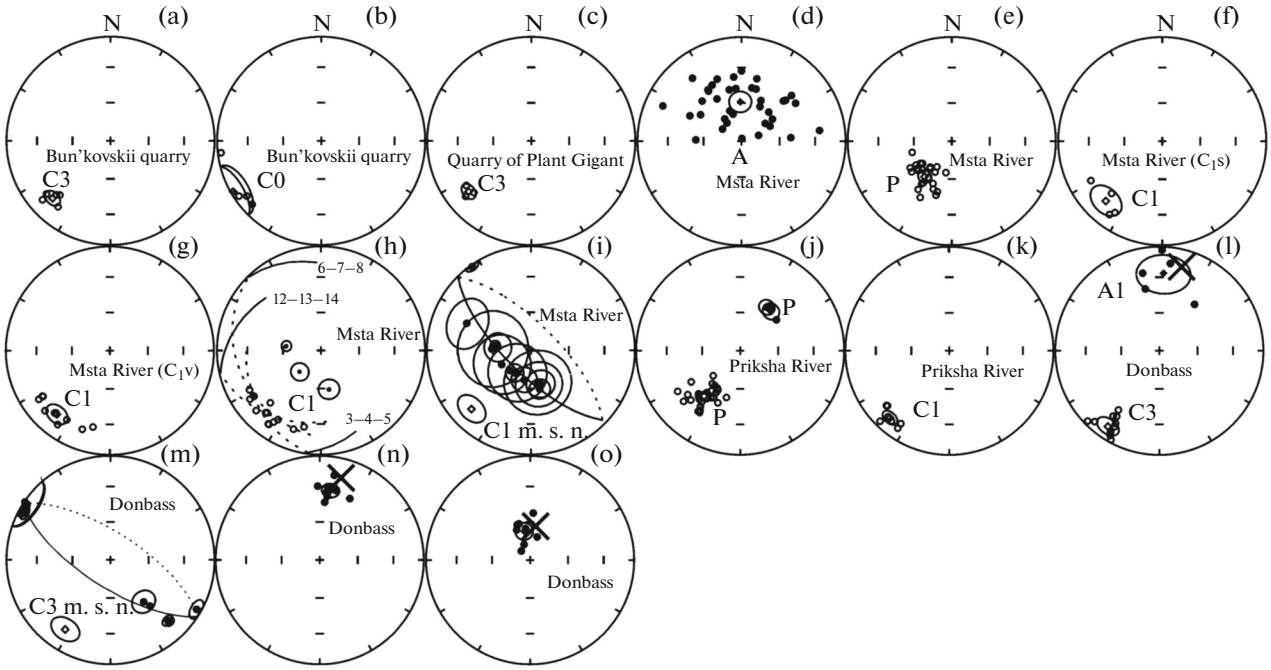


Fig. 7. Distribution of NRM components after demagnetization: (a), (b), Gzhelian rocks, Bun'kovskii quarry; (c) Kasimovian, Gigant quarry; (d), (e), (f), Serpukhovian, Msta River; (g), (h), (i), Visean, Msta River; (j), (k), Visean, Priksha River; (l), (m), (n), (o), basal part of Kartamyshain suite, Donbass ((l) components C3 and A1; (m), component C3 determined by m. s. n.; (n), (o) component A; (l), (m), (n), in stratigraphic coordinates; (o), in geographic coordinates; cross shows position of direction of present magnetic field in sampling region). Open and filled circles are vector projections on upper and lower hemispheres, respectively; distribution of component A is shown in geographic coordinates; distributions for other magnetization components J_n are shown in stratigraphic coordinates.

for the Visean rocks of the Priksha River is shown in Fig. 7j–7l, and their statistical characteristics are presented in Table 2.

The repeated studies of the collection of samples from the basal parts of the Kartamysh suite in Donbass established two J_n components. Components A and A1 are isolated in the temperature intervals 100–500 and 200–670°C. Some samples have both these components (Fig. 6, sample *5511AD*). The distributions of components A and A1 are shown in Figs. 7l, 7n, and 7o, and the statistical characteristics are presented in Table 2. The second component C3 is isolated in the temperature interval 200–670°C, Fig. 6, sample *5526U2*. The distribution of component C1 is not Fisher's (Fig. 7l). For the samples having component C1, we used the method of sliding normals to the remagnetization circles (Khramov et al., 2011) (Fig. 7m, Table 2).

For comparison, in Table 3 we present the results of the previous studies for the sections that are revisited in the present work. Table 4 summarizes the existing paleomagnetic determinations for the Cretaceous rocks of the EEP in the interval 300–344 Ma.

DISCUSSION

The Lower, Middle, and Upper Cretaceous formations of the Moscow syneclise, just as the other marine carbonate and terrigenous-carbonate Paleozoic deposits of the EEP, have a polycyclic structure (Fig. 2) resulting from the eustatic (global) sea-level fluctuations against the background regional tectonic motions (Makhlina and Shik, 1983; Makhlina et al., 2001a; Tikhomirov, 1968; Alekseev, 1996). According to the published litho-facial reconstructions (Osipova and Vel'skaya, 1967; Makhlina et al., 1993; 2001a; 2001b), the eustatic fluctuations resulted in the fact that the spatial positions of a number of facies changed many times throughout the entire Cretaceous. These facies are the peripheral facies, shelf, and marginal lagoons; shallow coastal water; shallow open sea; and relatively deep-water marine facies. The regime of stable subsidence has been repeatedly changed by the overall shallowing and contraction of the sedimentary basin. This situation inevitably produces favorable conditions for the remagnetization (additional magnetization) of the rocks. For instance, this is the case with the aridization of the climate, which creates the possibility for chemical processes. This is also fixed by the dolomitization of the limestones. The subsequent existence of the geomagnetic field of reversed polarity for 60 Ma (Kiama hyperchron), together with the changes of the paleogeographic settings, has even further facilitated the remagnetization of the Cretaceous deposits and formation of the J_n components of reversed polarity.

Table 3. Average paleomagnetic poles for studied Carboniferous deposits of Russian platform

No.	GPMDB result no./ source	φ,° N	λ,° E	Φ,° N	Λ,° E	dp°	dm°	Age, Ma
Gzhelian								
1	4900	55.6	38.5	42	167	1	2	303
2	4899	55.6	38.5	46	177	3	5	303
3	4901	56	38.5	42	169	1	2	303
4	/(1)	55.6	38.5	43	167	3	5	303
5	t. w.	55.3	38.6	34	162	4	7	
6	C_3g		$N = 5$	42	168	$A_{95} = 5°$		303
Moskovian								
7	/(1)	56.2	34.4	41	166	4	7	312
8	4913	56.5	34.5	33	159	2	4	313
9	/(1)	54.9	37.4	38	166	4	6	315
10	C_2m		$N = 3$	37	164	$A_{95} = 5°$		313
Lower Carboniferous (C_1s–C_1v), component P								
11	/(1)	59.5	33.9	52	151	4	6	
12	4938	59.5	34.0	49	164	6	4	
13	t. w.	58.3	34.0	56	152	4	6	
14	t. w.	58.3	33.6	53	152	9	14	
15	/(2)	58.3	33.6	52	164	3	5	
16	4937	59.3	34.0	45	156	3	5	
17	t. w.	58.3	33.6	47	158	3	5	
18	P		$N = 7$	51	157	$A_{95} = 3°$		265–270
Lower Carboniferous (C_1s–C_1v), component C1								
19	t. w.	58.3	34.0	31	160	6	12	326
20	t. w.	58.3	33.6	30	164	3	6	335–345?
21	/[1]	59.5	33.9	27	159	5	9	335–345?
22	C_1v (20 + 21)	$N = 2$		29	159	$A_{95} = 2°$		335–345?

φ,° N, λ,° E are sampling coordinates; Φ°, Λ° are latitude and longitude of paleomagnetic pole; dp°, dm° are semiaxes of confidence oval; A°_{95} is radius of confidence at 95% probability for pole's direction; determination indices according to GPMDB (McElhinny and Lock, 1996; Pisarevsky, 2005); (1) stands for (Iosifidi and Mikhailova, 2017); (2) stands for (Pogarskaya, 1984); t. w. stands for this work; * indicates that data are averaged through the R-resultant-vector of NRM.

According to the existing magnetostratigraphic schemes (*Dopolneniya*..., 2000; *The Geologic*..., 2012), the Cretaceous (Visean–Gzhelian) was dominated by a geomagnetic field with frequently changing polarity (the intervals between the reversals were about 1 Ma). In this case, the simultaneous remagnetization of all the studied sections in the Cretaceous by the field of reversed polarity is barely possible. Hence, the interpretation of the data should allow for the probable remagnetization both in the Cretaceous and during the Kiama hyperchron.

The obtained paleomagnetic directions for the Cretaceous rocks of the Bun'kovskii quarry section and the Gigant plant quarry section have reasonable convergence by the C3 components with the results of the first studies of the discussed collections (Table 3). The presence of the C0 component which, to all appearances, is associated with the shallowing of the magnetic inclinations due to the aggregation of the particles by the interaction of the magnetic and clay grains (Bagina, 1966; 1968; Iosifidi and Mikhailova, 2017) supports the absence of the full remagnetization of the rocks. In the first studies of the Gzhelian clays (Shchelkovskaya stratum) in the paleomagnetic laboratory of VNIGRI, Komissarova carried out experiments on the redeposition of the studied samples. The presoaked specimens were redeposited in a water column ($10 \times 10 \times 55$ cm) by batchwise filling (the labo-

Table 4. Paleomagnetic poles for Carboniferous deposits of EEP in 300–344 Ma interval

No.	GPMDB result no./ source	Rock type	Sampling coordinates		*Q*	Pole coordinates		dp°	dm°	Age, Ma
			φ,° N	λ,° E		Φ,° N	Λ,° E			
1	2	3	4	5	6	7	8	9	10	13
1	5823	Dolerites	55.5	13.5	4	38.5	168.5	3.4	6.7	300
2	/(3)	Limestones, sandstones, red rocks	48.5	37.8	6	42.0	179.0	2.0	4.0	302.5
3	/(4)	Limestones, sandstones, red rocks	48.3	38.3	5	44.0	167.0	1.0	2.0	303
4	3169	Igneous rocks	50.5	16.5	3	43.0	174.0	7.0	13.8	303
5		$N = 4$				42	172	$A_{95} = 4°$		302
6	3240	Sediments, red rocks	50.0	15.0	5	38.4	162.7	8.2	8.2	305
7	3333	Sediments, red rocks	50.2	14.0	4	35.5	153.8	1.3	2.6	308.5
8	/(4)	Limestones, sandstones, red rocks	48.8	38.6	5	35.0	162.0	2.0	3.0	314.5
9		$N = 3$				36	160	$A_{95} = 4°$		309
10	3332	Sediments, red rocks	49.8	13.3	4	31.1	151.7	2.6	5.1	318
11	5814	Sediments	59.7	10.4	4	39.9	159.9	5.0	9.4	329
12		$*N/n = 2/104$				34	155	$A_{95} = 4°$		324
13	2927	Igneous rocks sediments	51.5	11.5	3	29.8	153.7	12.2	12.2	338
14	8961	Dolerites	71.0	25.7	5	14.4	140.1	4.4	8.6	344

Q is reliability criterion of paleomagnetic data (Van der Voo, 1990); (3) stands for (Iosifidi et al., 2010); (4) stands for (Meijers et al., 2010); other denotations are described in notes to Table 3.

ratory field had an inclination $I = 70°$). Each new batch was filled after clearing the water column. In the case of the redeposition of whole specimens (5 × 5 × 5 cm), the average inclination was $I = 63° \pm 12°$ ($n = 8$), and for the 2.5 × 2.5 × 2.5 cm specimens, $I = 68° \pm 3°$, which supports the possibility of the inclination shallowing due to the formation (given the large amount of clay material) of particle aggregations due to the interaction of the magnetic and clay particles.

The C3 component which is established in the Gzhelian and Kasimovian rocks is likely to be the superposition of the primary and secondary (chemical) magnetizations J_n. This is indicated by the highly precise parameters of the revealed components of J_n and the presence of the C0 component. In the case of the purely chemical nature of J_n, the particles carrying the C0 component would also have been remagnetized. The reasonably good convergence, in terms of the C3 components, with the existing data for the EEP (Iosifidi and Mikhailova, 2017) places the time interval of additional magnetization close to the time of the formation of the rocks. This is also supported by the positive test of the consistence between the average direction for the Gzhelian rocks (Table 3, line 6) and the existing data for the EEP (Table 4, line 5), class B, $\gamma/\gamma_{cr} = 3°/7°$. The paleomagnetic determinations from the Moskovian rocks (Table 3, lines 7–10) obtained in (Iosifidi and Mikhailova, 2017) also pass the test for consistence with the other data for the EEP (Table 4, line 9), class B, $\gamma/\gamma_{cr} = 3°/8°$.

In the rocks of the Serpukhovian and Visean age, besides the secondary components A and P (Table 2),

the analysis also established the presence of the C1 component. In the rocks of the Msta River, the component C1 is identified from clays and limestones, which reduces the probability of the relation of this component to inclination shallowing. The method of sliding normals to the remagnetization circles (Khramov et al., 2011) yields a result close to the coeval poles for the EEP (Iosifidi and Mikhailova, 2017). The consistency test of the obtained average direction for the Serpukhovian/Visean boundary (Table 3, line 19) and the existing data for the EEP (Table 4, line 12) is positive, class C, $\gamma/\gamma_{cr} = 5°/12°$.

The magnetomineralogical studies in the rocks of the Msta River have shown the presence of fine-grain hematite which was formed at the latest stages of rock formation. All the cited peculiarities of identifying component C1 in the rocks of the Msta River suggest its primary origin.

In the sections of the Priksha River and mine 13, in the town of Boksitogorsk, our analysis also established primary components—a P component with normal polarity and a P component with reversed polarity (from one stratigraphic level). The first studies of the collections also established the same components (Table 3) (Khramov et al., 1974; Pogarskaya, 1984). The component P was probably formed at the end of the Kiama hyperchron. The reversal test for component P is positive (Table 3, lines 11–12), class B, $\gamma/\gamma_{cr} = 8°/9°$.

In the rocks of the Tul'skii horizon of the Visean Stage of the Carboniferous (a section of mine 13, the town of Boksitogorsk, and Priksha River), the C1 component was revealed for the first time. This component has a high precision parameter and, to all appearances, just as component C3, has a secondary chemical component J_n. The difference lies in the fact that component C3 reasonably well agrees with the existing data for the EEP (Iosifidi and Mikhailova, 2017), whereas the component C1 established from the rocks of the Tul'skii horizon of the Visean Stage is close that for the younger deposits (Early Serpukhovian–Late Visean) of the Msta River. The rather scarce determinations for the EEP in the interval 335–359 Ma (Table 4, lines 13, 14) are contradictory, which prevents us from classifying the C1 component for the sections of mine 13, town of Boksitogorsk, and Priksha River as primary. Judging by the positions of the paleomagnetic poles, component C1 in the sections of mine 13, town of Boksitogorsk, and Priksha River was formed in the Late Visean. There are insufficient data for the Lower Carboniferous and the determinations of the paleomagnetic directions from the bipolar J_n components for the correct estimation of the obtained results. These data would increase the reliability of the results.

The studies of the collection of the Late Carboniferous Donbass rocks were intended for retrieving the bipolar component J_n. This would have raised the status of the average paleomagnetic pole for the Gzhelian Stage of the Carboniferous. However, the repeated investigation of the samples from the basal parts of the Kartamysh suite of Donbass has not supported the presence of the Kartamyshian magnetic event (Davydov and Khramov, 1991; *A Geological...*, 2004). The obtained results show that the distributions of the components with positive polarity A and A1 are close to the present geomagnetic field in the sampling region in both coordinate systems (Figs. 7n, 7l, 7o; Table 2). Component C3 has a higher precision parameter in the stratigraphic coordinates than in the geographic coordinates, and the position of the paleomagnetic pole is close to the data for the Gzhelian rocks (Table 3). However, the results of the method of sliding normals to the remagnetization circles (Khramov et al., 2011) shows the lower determination error of the average direction in the geographic (present-day) coordinates (Table 2). This testifies to the presence of the secondary (Permian) component, which was not removed by the thermal demagnetization. Unfortunately, we only had a part of the samples of the collection from the basal layers of the Kartamysh suite of Donbass at our disposal and the statistics are not very high; nevertheless, the obtained results do not support the existence of the Kartamyshian event in the Gzhelian.

All the established secondary J_n components which differ from the present direction of the geomagnetic field have one feature in common: their directions correspond to the geomagnetic field of reversed polarity. With the closeness of the Kiama hyperchron taken into account, this can indicate the probable Late Paleozoic remagnetization of a part of the rocks. Our results support this. The revealed secondary components P (Table 2) have a high precision parameter, which is due to the formation of a secondary chemical magnetization.

The positions of the average paleomagnetic poles for the Gzhelian, Moskovian, Early Serpukhovian, and Late Visean rocks form a consistent sequence and agree with the existing data for the EEP, which was also shown at the first stage of the work (Iosifidi and Mikhailova, 2017). According to the obtained data (Table 3), the paleolatitudinal position of the EEP on the interval from 326 to 303 Ma varied from 11° N to 17° N (the latitudinal displacements were calculated the latitudinal displacements were calculated by coordinates 57° N, 35 ° E).

Thus, the conducted studies outlined the direction of the future work on revisiting the old collections of the Cretaceous deposits of the Russian platform. This is primarily a reinvestigation of the collections in which the first researchers identified the bipolar J_n components and the samples are not exclusively red rocks.

ACKNOWLEDGMENTS

We thank our teacher A.N. Khramov (1927–2014) for his constant interest and constructive comments in discussing the paleomagnetic studies on the Russian platform. We are also grateful to A.N. Didenko and V.E. Pavlov for their constructive criticism and comments.

The studies were partly conducted with the use of the facilities of the GEOMODEL and Nanotechnology Resource Centers of the Research Park of St. Petersburg University.

The work was supported by the Russian Foundation for Basic Research (project no. 16-05-00603a).

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Translated by M. Nazarenko